Centre-of-Expansion shell cosmology: from Migkas anisotropy to a flat CMB

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with computational assistance (Grok Bot / Cosmology)

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Big-picture letter: Migkas directional Hubble data fix an off-centre Sedov–Taylor / LTB shell geometry; the same geometry is then tested against the CMB. The CMB is an external test, not a fit target.

1 Problem and stance

Migkas et al. report sky-dependent cluster Hubble rates. We take that anisotropy seriously *and* take CMB isotropy seriously. Goal: one geometry that explains both.

2 Workflow

1. Migkas $A/A_{\rm all}$ vs. Galactic l (and Fig. 8); slice fits by redshift.
2. Fit CoE + thick ordinary-matter shell (Newtonian and relativistic LTB).
3. Orient sky: untilted → +15° tilt → +80.8° roll; overlay on Fig. 8 with solid MAX/MIN bands (no loop opacity gradient).
4. Donut-universe evolution: shells $> c$, visible sphere at c, recombination front marked.
5. CMB stress test: gravitational multipole → kill function → two-emitter construction.

Retired: Amp× sin panel with connector lines between amplitude and residuals—never reproduce it.

3 Best-fit geometry

Local Hubble velocity is free-fall relative to the CoE minus our motion, $\vec{V}_H = \vec{V}_g - \vec{V}_O$. We sit inside a thick shell; vision radius $R \approx 14.3\,$Gpc. Headline relativistic picture: $D \approx 32.9\,$Gpc, $R \approx 14.3\,$Gpc, $R_{\rm in} \approx 18.1\,$Gpc, $R_{\rm out} \approx 49.7\,$Gpc.

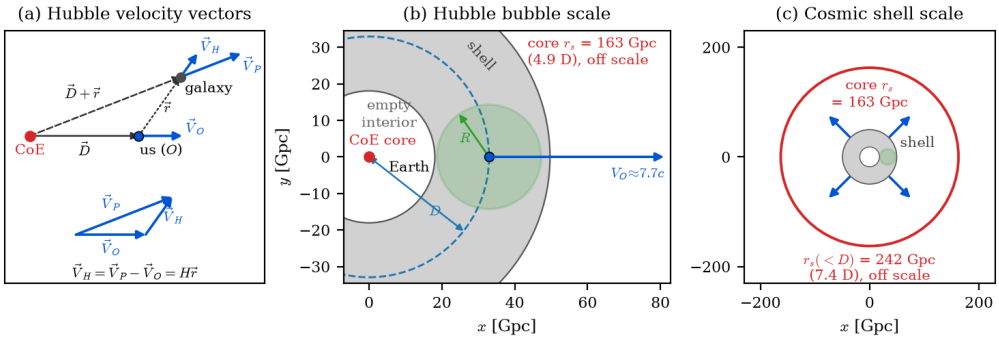


Figure 1: Geometry: CoE at left, observer (US) inside a thick shell, vision sphere of radius R. Local H is the difference of free-fall velocities relative to the CoE—not expanding-space redshift.

Table 1: Newtonian vs. relativistic shell+core fits to Migkas $A(l)$. Relativistic LTB with $\Omega_s = 0.049$ is the headline working model.

	Newt. Ω_s free	Newt. $\Omega_s = 0.049$	Rel. LTB $\Omega_s = 0.049$
D (Gpc)	93.7	49.2	32.9
Ω_s	1.01	0.049	0.049
l_0 (deg)	12.8	13.0	13.5
R^2 vs Migkas	0.909	0.893	0.885
R^2 vs A_F	0.987	0.977	0.963

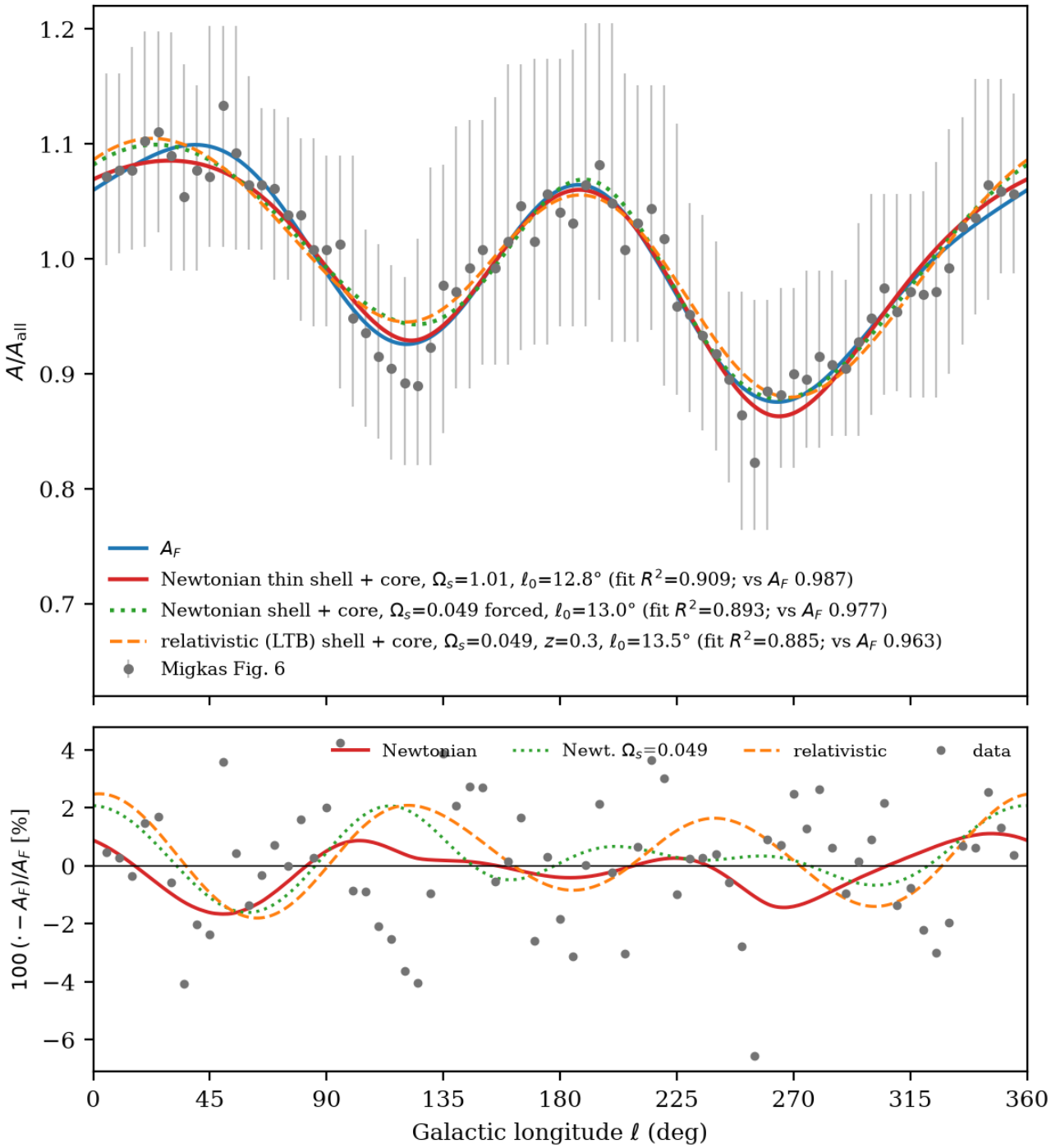


Figure 2: Migkas Fig. 6 $A/A_{\rm all}$ vs. Galactic l. Grey = data; red/green/orange = Newtonian (free Ω_s), Newtonian ($\Omega_s = 0.049$), relativistic LTB ($\Omega_s = 0.049$). Lower panel: percent difference from A_F. No connector-line figure.

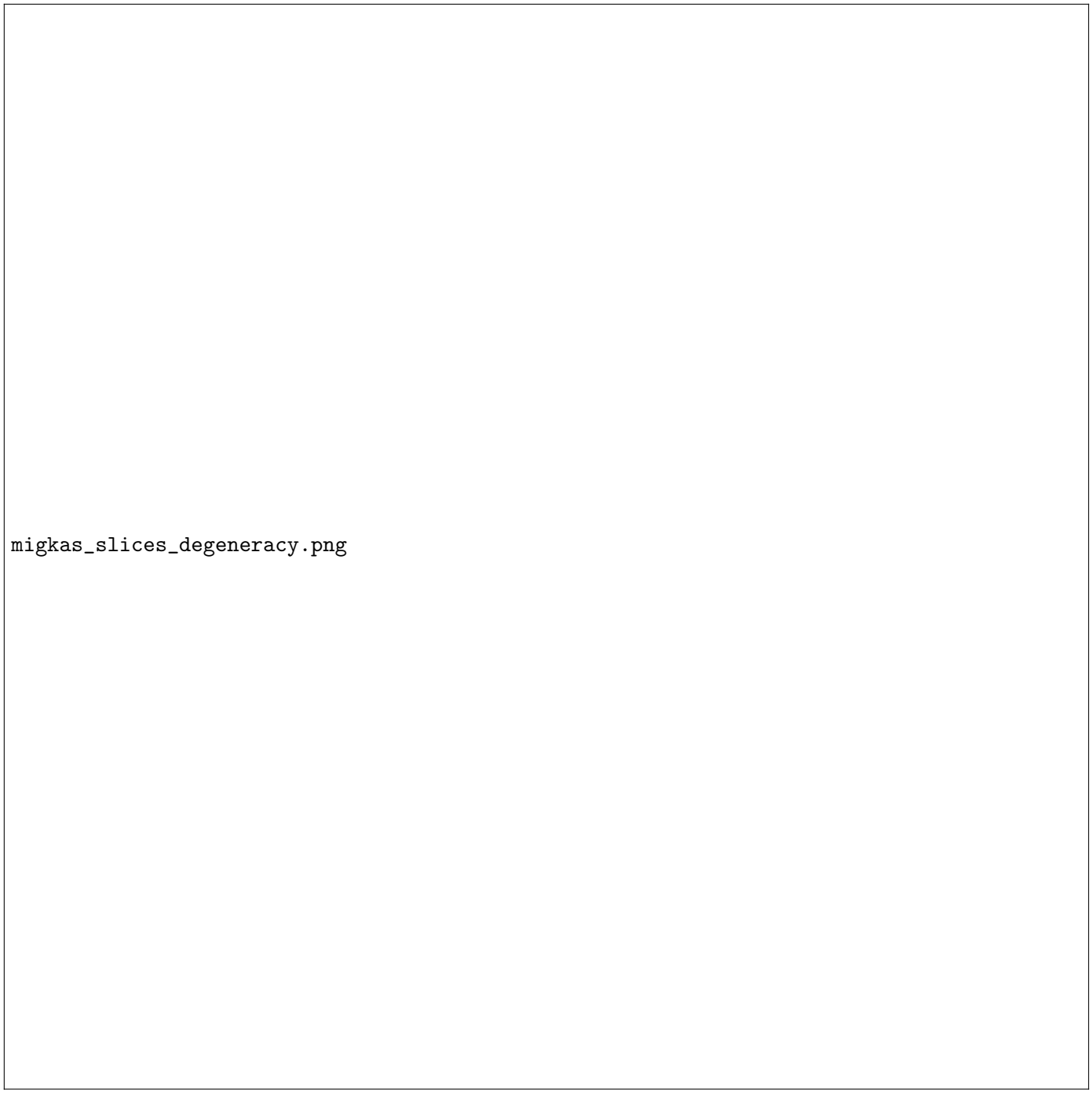


Figure 3: Left three: CoE model recomputed on Migkas redshift slices (0.03–0.07, $z > 0.07$, all-z); extrema move with depth. Right: D–mass degeneracy—any D fits if core mass scales; data constrain the core's share of critical density, not D alone.

4 Money shot: sky orientation

Each panel below is one full three-panel stage (model frame / Mollweide / 3D), stacked full-width—never side-by-side. Sequence: untilted → +15° tilt → tilt+80.8° roll with solid loops. Four MAX/MIN loops (±9°) track $A(l)$ extrema onto the sky.

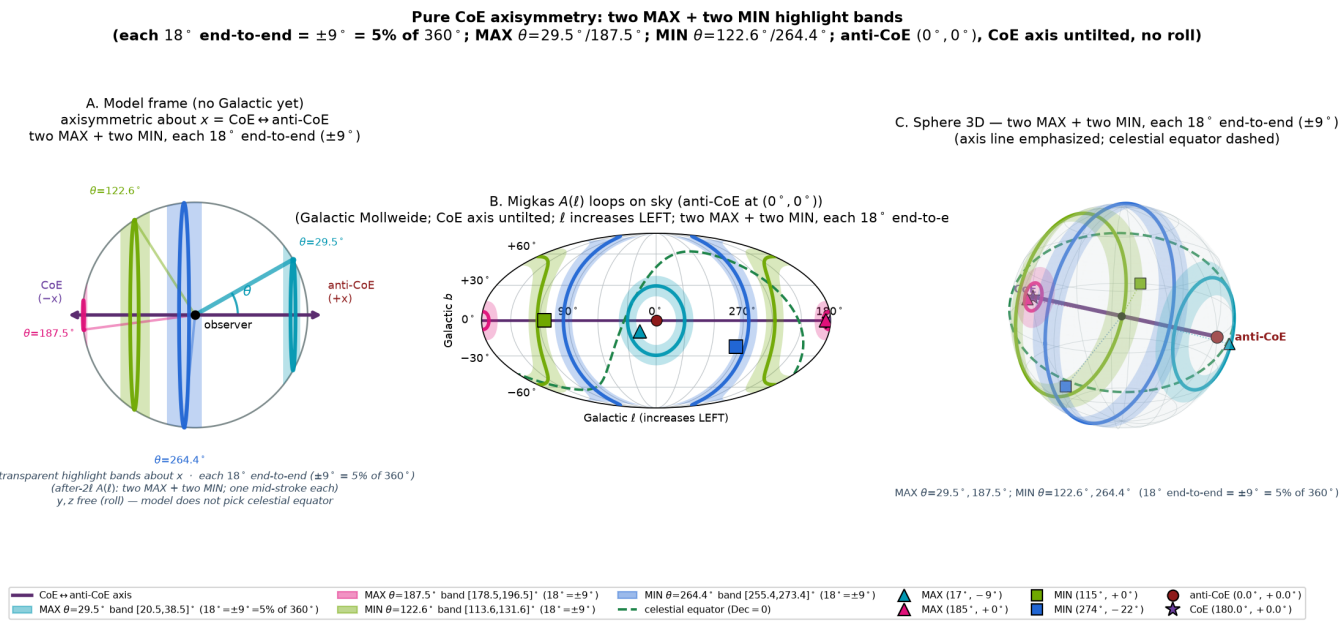


Figure 4: Stage 1 — untilted, no roll. Side view: CoE↔anti-CoE; Mollweide and 3D place the four loops on the celestial sphere.

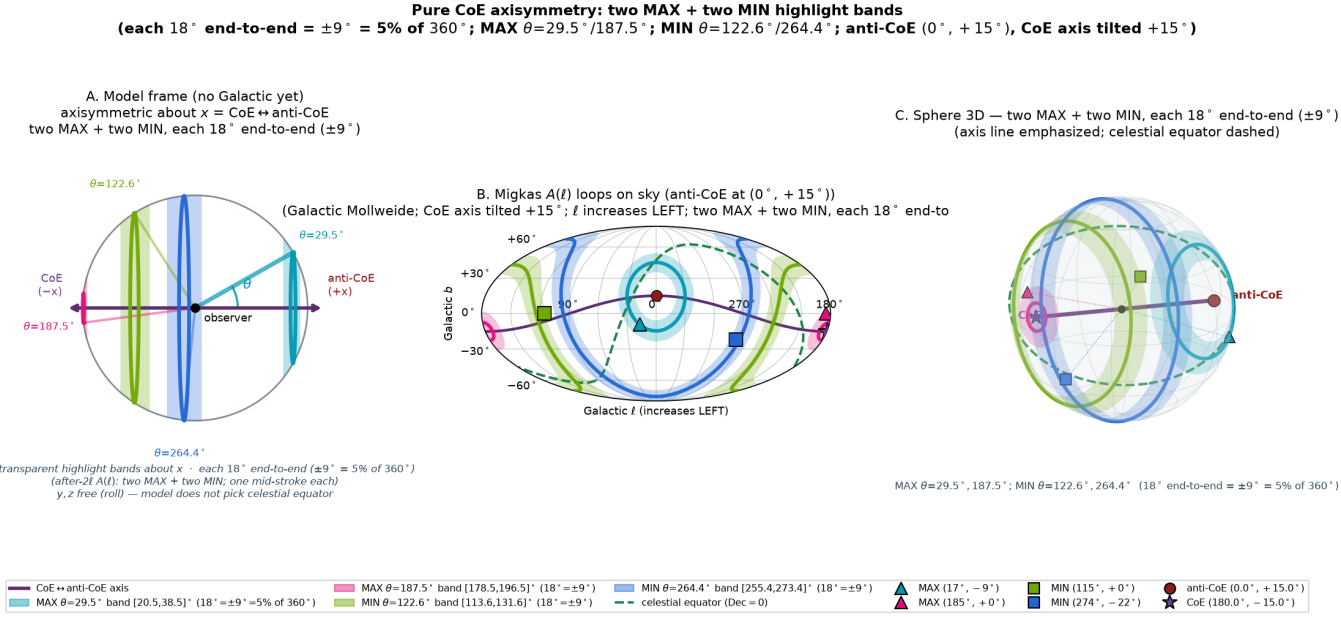


Figure 5: Stage 2 — +15° tilt of the CoE axis.

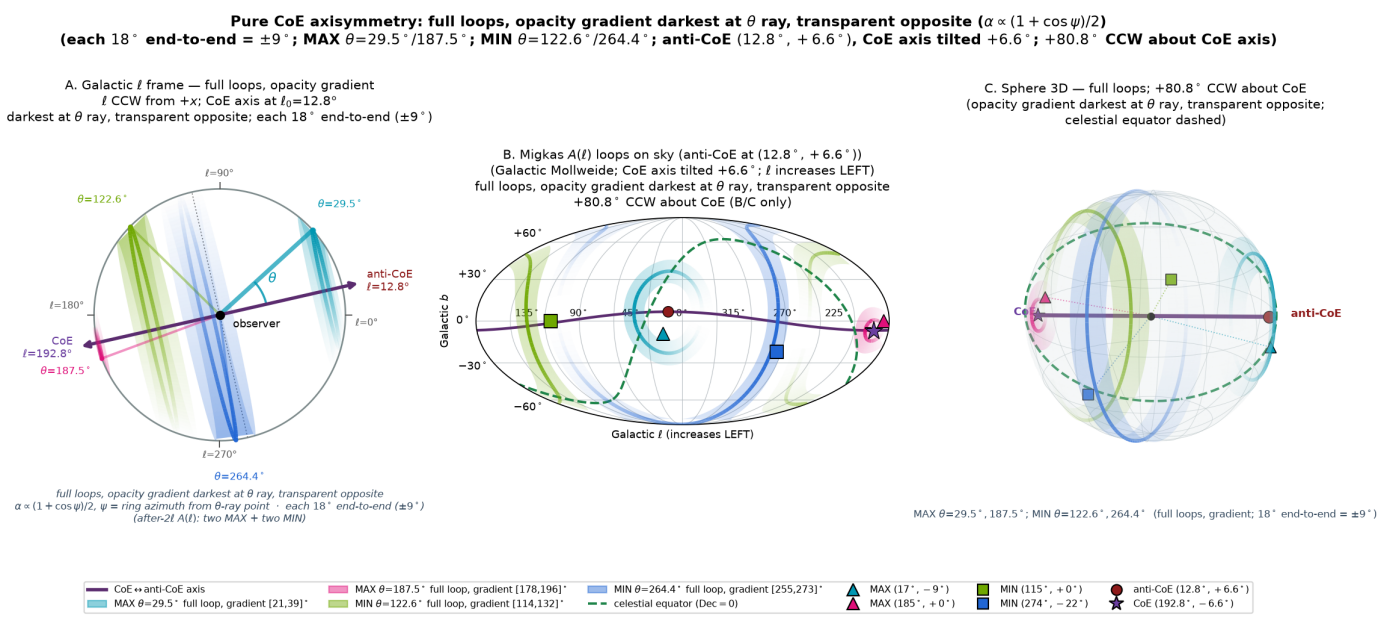


Figure 6: Stage 3 — +15° tilt + +80.8° Galactic roll, solid loops (no opacity gradient).

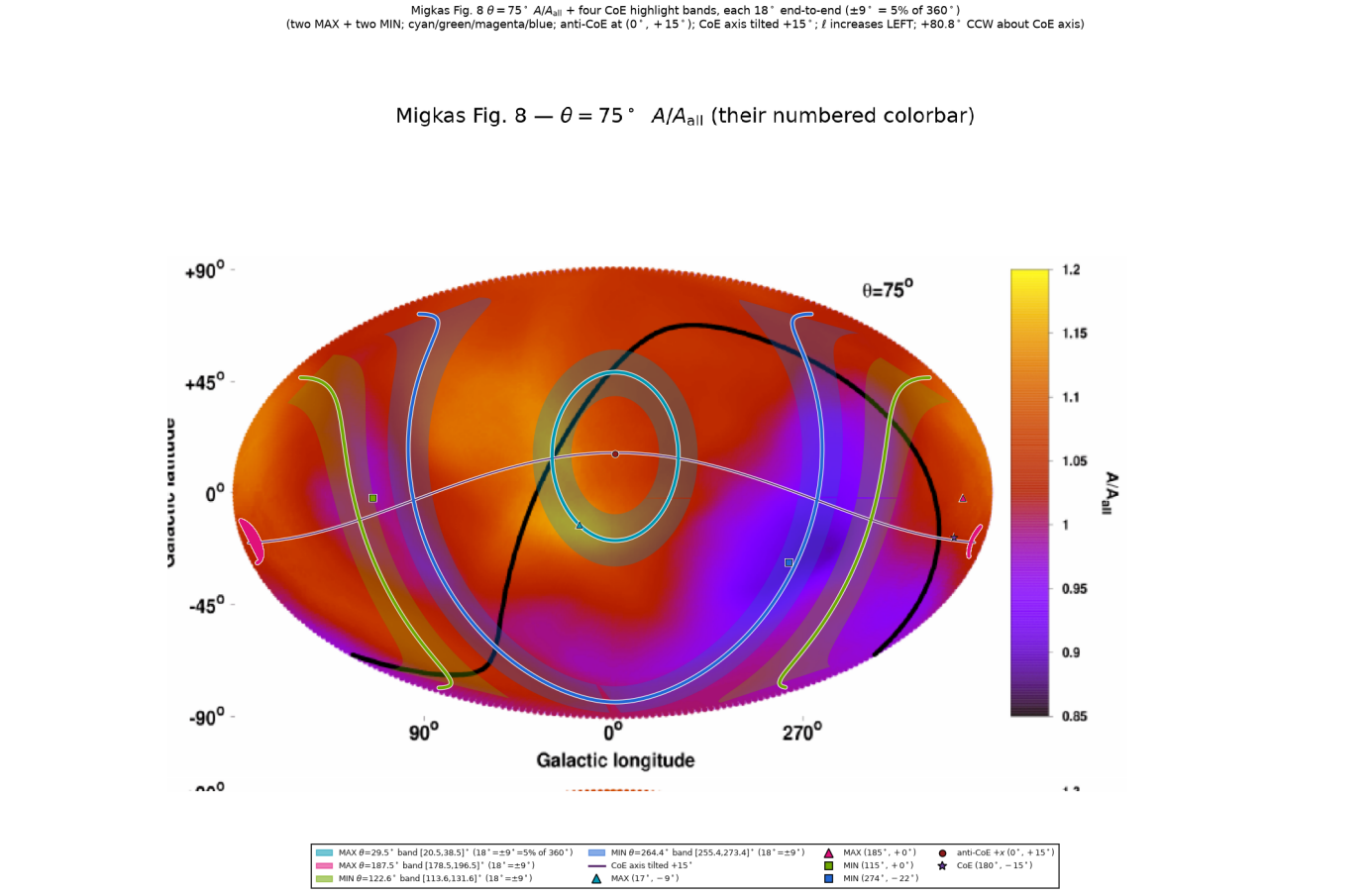


Figure 7: Migkas Fig. 8 overlay ($\theta = 75°$ $A/A_{\rm all}$): tilt + 80.8° roll, all four MAX/MIN loops as solid ±9° bands—no loop opacity gradient.

5 Time evolution and CMB path

The *donut universe*: CoE fixed; shells expand faster than light; the visible sphere around US grows at c; a CoE-centred recombination front expands with the shells. For the CMB construction, US sits between two emitters that share one bulk velocity vector—relative Doppler vanishes until gravity's multipole is added.

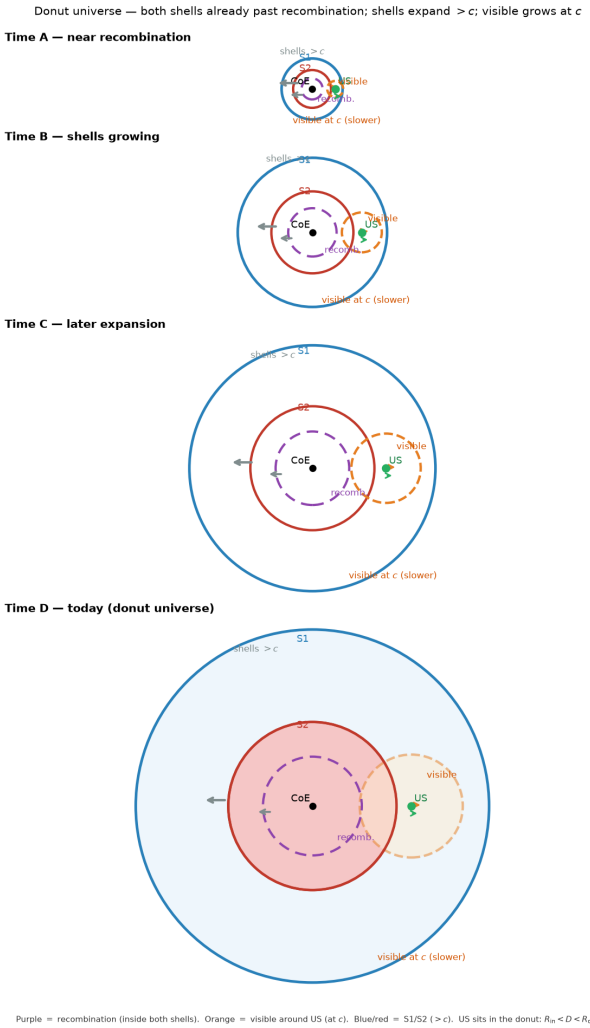


Figure 8: Donut universe (A→D): both shells already outside the recombination front; shells expand $> c$; visible sphere around US grows at c (slower).

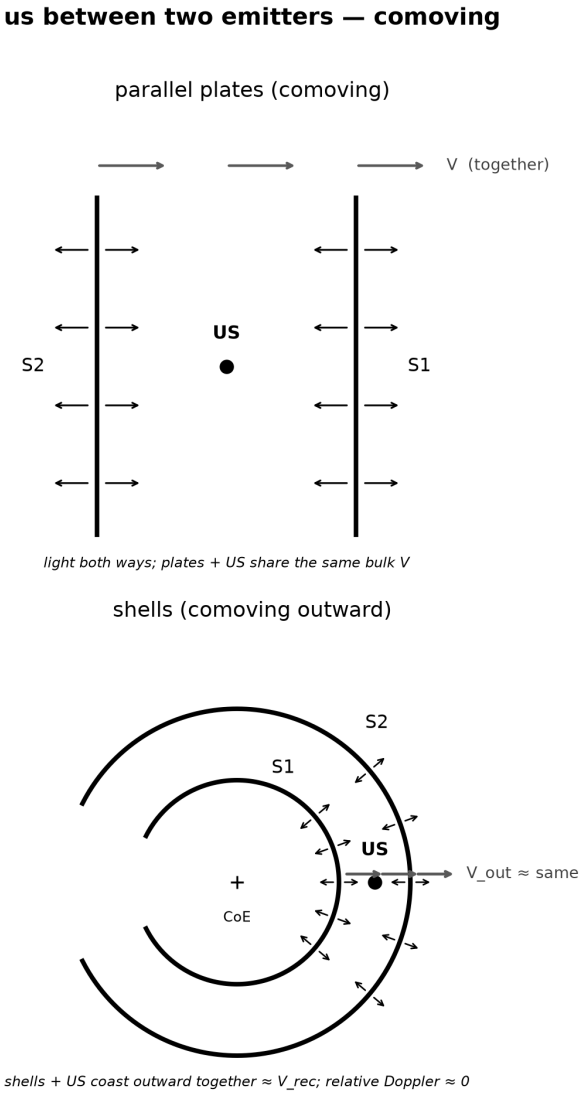


Figure 9: Two-emitter baseline: plates (top) or shells (bottom) share bulk $\mathbf{V}$ with US $\Rightarrow$ flat sky before the multipole.

The same off-centre geometry produces a gravitational multipole on the last-scattering sky. Left alone it would ruin CMB isotropy. We compute the Doppler residual on a full 0°–360° circle about the CoE axis and define a *required kill function* as its opposite—the fractional energy pattern that would cancel it. Fixed-$R = 14.3\,\mathrm{Gpc}$ Doppler amplitudes sit far above 10^{-5} unless cancelled. At recombination, $f = T_0/3000 \approx 9.08 \times 10^{-4}$, $R_{\mathrm{see}} \approx 4.25\,\mathrm{Gpc}$, and $\varepsilon_{\mathrm{rec}} = D_{\mathrm{rec}}/R_{\mathrm{see}}$ is only a few $\times 10^{-3}$ (nearly centred).

As an exploratory step we ask whether the two-emitter construction, with a multipole partner from the same physics, can match the kill function. Figure **??** compares unit-normalized shapes; at larger D the exploratory residuals track the kill morphology. Figure **??** subtracts a dipole+P_2 (and $\ell \leq 4$) partner from the CoE Doppler: residuals fall sharply—about 85% of demeaned power removed at $D = 49\,\mathrm{Gpc}$ for $\ell \leq 4$. Not yet 10^{-5}, but the construction produces the *kind* of multipole the kill function needs.

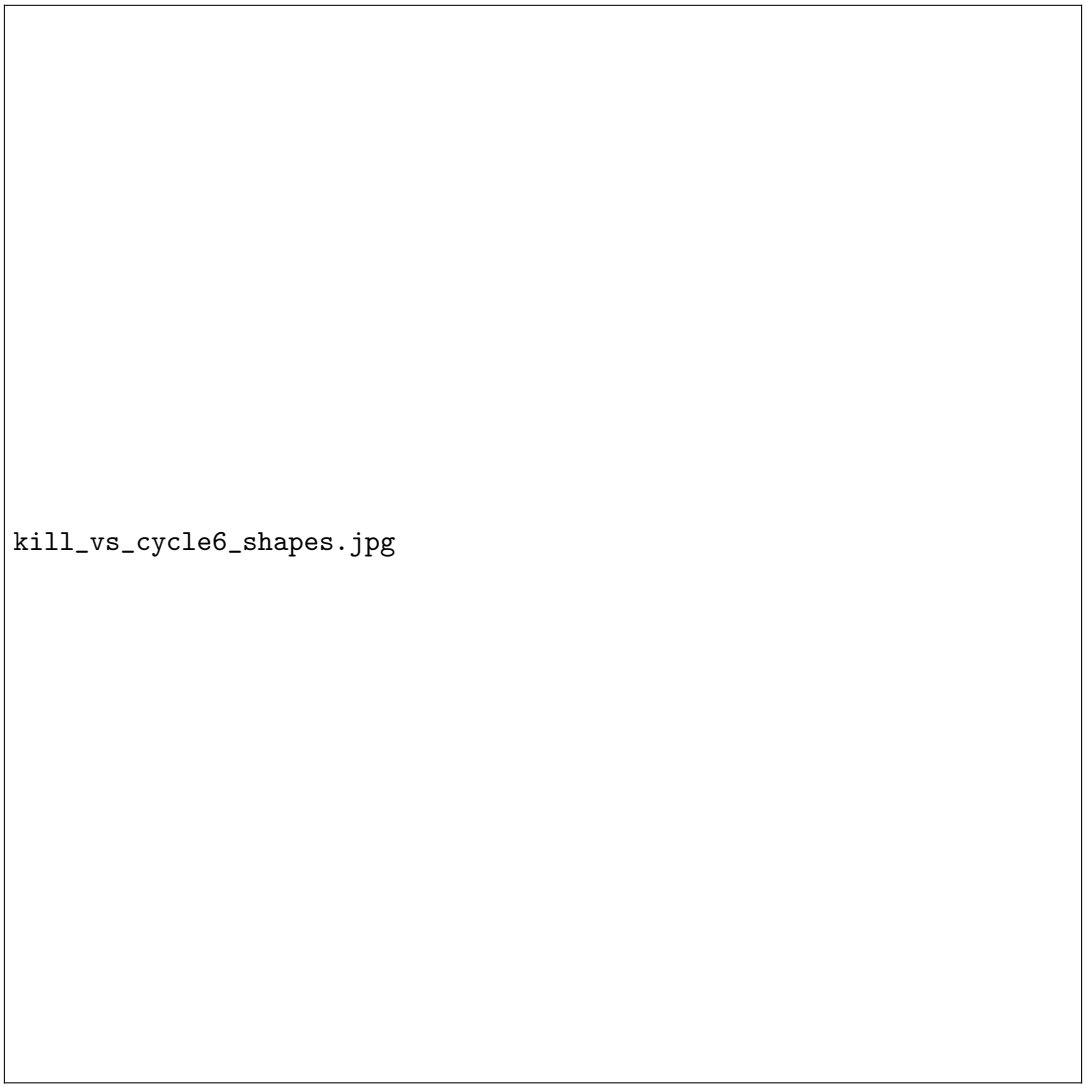


Figure 10: Unit-normalized shapes at three D: required kill ($-$Doppler, thick red) vs. exploratory two-emitter / shear residuals. Morphology converges at larger D.

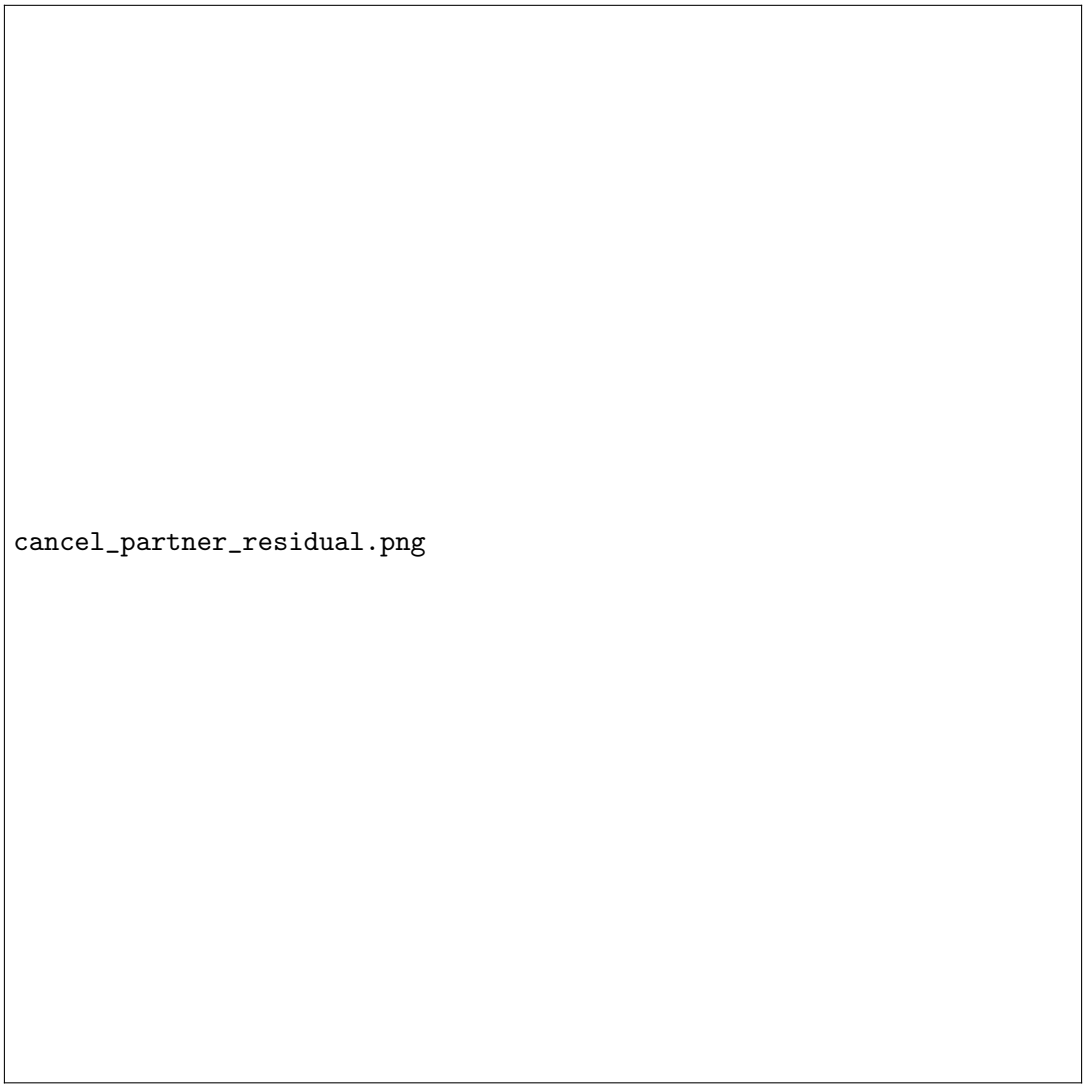


Figure 11: CoE Doppler plus opposite multipole partner (monopoles removed). Grey = Doppler; red = required kill; green/purple = residual after dip+P_2 and after $\ell \leq 4$. Residuals shrink toward zero, especially at $D = 49\,\mathrm{Gpc}$.

6 Open question

Where were we at recombination? Evolve $D(t)$, $R_{\mathrm{in}}(t)$, $R_{\mathrm{out}}(t)$; null geodesics $\rightarrow T(\theta)$; keep the Migkas quadrupole while driving $A_{\ell \geq 2} \lesssim 10^{-5}$.

7 Bottom line

Migkas-fixed CoE+shell fits the directional Hubble signal across redshift slices (with a D–mass degeneracy). The same geometry predicts a CMB-scale multipole we write as a kill target; a two-emitter construction tracks that target in shape and cancels most of it with a low-order partner. Small $\varepsilon_{\mathrm{rec}}$ helps. Much work left; not a dead end.

Respectfully,

Chris Alameddin